Session-9-exercises.R

```r
###########################
### Session 9 Exercises ###
###########################

# Exercise 9.1
####################
#                  #
#    Exercise a    #
#                  #
####################

text1 <- "The dragon year is 2025"

####################
#                  #
#    Exercise b    #
#                  #
####################

my_pattern <- "[0-9]"
grepl(my_pattern,text1)
```

```
## [1] TRUE
```

```r
####################
#                  #
#    Exercise c    #
#                  #
####################

string_position <- gregexpr(my_pattern,text1)
string_position[[1]][1:length(string_position[[1]])]
```

```
## [1] 20 21 22 23
```

```r
####################
#                  #
#    Exercise d    #
#                  #
####################

my_pattern <- "\\d[A-Z]"
grepl(my_pattern,text1)
```

```
## [1] FALSE
```

```r
####################
#                  #
#    Exercise e    #
#                  #
####################

my_pattern <- "\\s"
first_space <- regexpr(my_pattern,text1)
first_space[[1]][1]

## [1] 4

####################
#                  #
#    Exercise f    #
#                  #
####################

my_pattern <- "[a-z].\\d"
grepl(my_pattern,text1)

## [1] TRUE

####################
#                  #
#    Exercise g   #
#                  #
####################

string_pos2 <- gregexpr(my_pattern,text1)[[1]][1]
string_pos2

## [1] 18

####################
#                  #
#    Exercise h    #
#                  #
####################

my_pattern <- "\\s[a-z][a-z]\\s"
string_pos3 <- gregexpr(my_pattern,text1)[[1]][1]
string_pos3

## [1] 16
```

```r
####################
#                  #
#    Exercise i    #
#                  #
####################

text2 <- sub(my_pattern," is not ",text1)
text2

## [1] "The dragon year is not 2025"

####################
#                  #
#    Exercise j    #
#                  #
####################

my_pattern <- "\\d{4}$"
string_pos4 <- gregexpr(my_pattern,text2)[[1]][1]
string_pos4

## [1] 24

####################
#                  #
#    Exercise k   #
#                  #
####################

str_sub(text2,start = string_pos4,string_pos4+1)

## [1] "20"

########################################################################
```

```r
# Exercise 9.2
# Load the tidyverse and read in the Medicare payments dataset
library(tidyverse)
names <- c("DRG", "ProviderID", "Name", "Address", "City", "State", "ZIP",
"Region", "Discharges", "AverageCharges", "AverageTotalPayments",
"AverageMedicarePayments")
types = 'ccccccccinnn'
inpatient <- read_tsv('inpatient.tsv', col_names = names, skip=1, col_types =
types)

# Separate at the fourth position
inpatient_separate <- separate(inpatient,DRG,c('DRGcode','DRGdescription'),4)

# And take a look at the data now
glimpse(inpatient_separate)

## Rows: 163,065
## Columns: 13
## $ DRGcode                 <chr> "039 ", "039 ", "039 ", "039 ", "039 ", "039 "…
## $ DRGdescription          <chr> "- EXTRACRANIAL PROCEDURES W/O CC/MCC", "- EXT…
## $ ProviderID              <chr> "10001", "10005", "10006", "10011", "10016", "…
## $ Name                    <chr> "SOUTHEAST ALABAMA MEDICAL CENTER", "MARSHALL …
## $ Address                 <chr> "1108 ROSS CLARK CIRCLE", "2505 U S HIGHWAY 43…
## $ City                    <chr> "DOTHAN", "BOAZ", "FLORENCE", "BIRMINGHAM", "A…
## $ State                   <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL"…
## $ ZIP                     <chr> "36301", "35957", "35631", "35235", "35007", "…
## $ Region                  <chr> "AL - Dothan", "AL - Birmingham", "AL - Birmin…
## $ Discharges              <int> 91, 14, 24, 25, 18, 67, 51, 32, 135, 34, 14, 4…
## $ AverageCharges          <dbl> 32963.07, 15131.85, 37560.37, 13998.28, 31633.…
## $ AverageTotalPayments    <dbl> 5777.24, 5787.57, 5434.95, 5417.56, 5658.33, 6…
## $ AverageMedicarePayments <dbl> 4763.73, 4976.71, 4453.79, 4129.16, 4851.44, 5…

# Trim the DRGcode field
inpatient_separate <- inpatient_separate %>%
  mutate(DRGcode=str_trim(DRGcode))

glimpse(inpatient_separate)

## Rows: 163,065
## Columns: 13
## $ DRGcode                 <chr> "039", "039", "039", "039", "039", "039", "039…
## $ DRGdescription          <chr> "- EXTRACRANIAL PROCEDURES W/O CC/MCC", "- EXT…
## $ ProviderID              <chr> "10001", "10005", "10006", "10011", "10016", "…
## $ Name                    <chr> "SOUTHEAST ALABAMA MEDICAL CENTER", "MARSHALL …
## $ Address                 <chr> "1108 ROSS CLARK CIRCLE", "2505 U S HIGHWAY 43…
## $ City                    <chr> "DOTHAN", "BOAZ", "FLORENCE", "BIRMINGHAM", "A…
## $ State                   <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL"…
## $ ZIP                     <chr> "36301", "35957", "35631", "35235", "35007", "…
## $ Region                  <chr> "AL - Dothan", "AL - Birmingham", "AL - Birmin…
## $ Discharges              <int> 91, 14, 24, 25, 18, 67, 51, 32, 135, 34, 14, 4…
## $ AverageCharges          <dbl> 32963.07, 15131.85, 37560.37, 13998.28, 31633.…
## $ AverageTotalPayments    <dbl> 5777.24, 5787.57, 5434.95, 5417.56, 5658.33, 6…
## $ AverageMedicarePayments <dbl> 4763.73, 4976.71, 4453.79, 4129.16, 4851.44, 5…
```

```r
# The DRGdescription field has a hyphen in front so we need to do something different
inpatient_separate <- inpatient_separate %>%
  mutate(DRGdescription=str_sub(DRGdescription, 3))

glimpse(inpatient_separate)

## Rows: 163,065
## Columns: 13
## $ DRGcode                  <chr> "039", "039", "039", "039", "039", "039", "039…
## $ DRGdescription           <chr> "EXTRACRANIAL PROCEDURES W/O CC/MCC", "EXTRACR…
## $ ProviderID               <chr> "10001", "10005", "10006", "10011", "10016", "…
## $ Name                     <chr> "SOUTHEAST ALABAMA MEDICAL CENTER", "MARSHALL …
## $ Address                  <chr> "1108 ROSS CLARK CIRCLE", "2505 U S HIGHWAY 43…
## $ City                     <chr> "DOTHAN", "BOAZ", "FLORENCE", "BIRMINGHAM", "A…
## $ State                    <chr> "AL", "AL", "AL", "AL", "AL", "AL", "AL", "AL"…
## $ ZIP                      <chr> "36301", "35957", "35631", "35235", "35007", "…
## $ Region                   <chr> "AL - Dothan", "AL - Birmingham", "AL - Birmin…
## $ Discharges               <int> 91, 14, 24, 25, 18, 67, 51, 32, 135, 34, 14, 4…
## $ AverageCharges           <dbl> 32963.07, 15131.85, 37560.37, 13998.28, 31633.…
## $ AverageTotalPayments     <dbl> 5777.24, 5787.57, 5434.95, 5417.56, 5658.33, 6…
## $ AverageMedicarePayments <dbl> 4763.73, 4976.71, 4453.79, 4129.16, 4851.44, 5…

###############################################################################

#9.3
# The simple version when fwf is working
read_csv("employees_broken.txt")

## # A tibble: 7 × 1
##   `EMPLOYEE DATA EXPORT – INTERNAL USE ONLY`
##   <chr>
## 1 Generated on: 2023-11-15
## 2 ----------------------------------------
## 3 001John Smith   19850423M  45000
## 4 002Alice Brown  19911205F  52000
## 5 003Mark  Davis  19790612M
## 6 004SophieMartin 19880228F  61000
## 7 005Paul  Durand 19930517 M 48000

# It seems that the first three lines are just headers and/or metadata
# No clear delimiter, but it seems to be fiwed width.

#c
employees_raw <- read_fwf(
  "employees_broken.txt",
  fwf_positions(
    start = c(1, 4, 17, 25, 28),
    end   = c(3, 16, 24, 26, 33),
    col_names = c("id", "name", "birthdate", "gender", "salary")
  ),
  skip = 3, na = c("", " "))
```

```
employees_raw

## # A tibble: 5 × 5
##   id    name         birthdate gender salary
##   <chr> <chr>            <dbl> <chr>   <dbl>
## 1 001   John Smith    19850423 M       45000
## 2 002   Alice Brown   19911205 F       52000
## 3 003   Mark  Davis   19790612 M          NA
## 4 004   SophieMartin  19880228 F       61000
## 5 005   Paul  Durand  19930517 M       48000

#d
# Name is in one column, should be separated into first and last name
# Birthdate is in numeric format, it should be date
# Salary has a missing value

# The actual problem
#e
read_csv("employees_broken2.txt")

## # A tibble: 7 × 1
##   `EMPLOYEE DATA EXPORT – INTERNAL USE ONLY`
##   <chr>
## 1 Generated on: 2023-11-15
## 2 ---------------------------------------
## 3 001John Smith   19850423M  45000
## 4 002Alice Brown  19911205F  52000
## 5 003Mark  Davis  19790612M
## 6 004SophieMartin19880228F  61000
## 7 005Paul  Durand 19930517 M 48000

# It is a seemingly fixed width file, but in fact, at least SophieMartin's
record is different. It can be solved either manually or reading the file
line-by-line as an ETL solution. You cannot “fix” this with clever column
positions.

# To check the file line by line
lines <- read_lines("employees_broken2.txt")
data_lines <- lines %>% discard(~ str_detect(.x, "^EMPLOYEE|^Generated|^-
{3,}"))
data_lines

## [1] "001John Smith   19850423M  45000" "002Alice Brown  19911205F  52000"
## [3] "003Mark  Davis  19790612M       " "004SophieMartin19880228F  61000"
## [5] "005Paul  Durand 19930517 M 48000"

# Observed invariant rules
# From inspection:
# • id = first 3 digits
# • birthdate = exactly 8 consecutive digits
# • gender = single letter M or F
# • salary = trailing digits (optional)
# • name = whatever remains between id and birthdate
```

```r
# These rules hold even when spacing shifts.

employees_parsed <- tibble(raw = data_lines) %>%
  mutate(
    id = str_extract(raw, "^\\d{3}"),
    birthdate = str_extract(raw, "\\d{8}"),
    gender = str_extract(raw, "[MF](?=\\s|\\d|$)"),
    salary = str_extract(raw, "\\d+$"),
    name = raw %>%
      str_remove("^\\d{3}") %>%       # remove id
      str_remove("\\d{8}.*$") %>%     # remove birthdate and everything after
      str_trim()
  ) %>%
  select(id, name, birthdate, gender, salary)

employees_parsed

## # A tibble: 5 × 5
##   id    name         birthdate gender salary
##   <chr> <chr>        <chr>     <chr>  <chr>
## 1 001   John Smith   19850423  M      45000
## 2 002   Alice Brown  19911205  F      52000
## 3 003   Mark  Davis  19790612  M      <NA>
## 4 004   SophieMartin 19880228  F      61000
## 5 005   Paul  Durand 19930517  M      48000

employees_clean <- employees_parsed %>%
  mutate(
    birthdate = as.Date(birthdate, format = "%Y%m%d"),
    salary = as.numeric(salary),
    gender = factor(gender, levels = c("M", "F"))
  )

employees_clean

## # A tibble: 5 × 5
##   id    name         birthdate  gender salary
##   <chr> <chr>        <date>     <fct>   <dbl>
## 1 001   John Smith   1985-04-23 M       45000
## 2 002   Alice Brown  1991-12-05 F       52000
## 3 003   Mark  Davis  1979-06-12 M          NA
## 4 004   SophieMartin 1988-02-28 F       61000
## 5 005   Paul  Durand 1993-05-17 M       48000

# The meta-lesson you take away
# - Data formats are contracts, not guarantees
# - Tools enforce assumptions — they do not verify them
# - The analyst is responsible for detecting violations
# - Sometimes the “right” solution is to abandon the format itself
# These are professional-grade insights, even if the code remains simple.
```

```r
# The first name - last name separation does not work simply by a space:
employees_clean_NW <- employees_clean %>%
  separate(
    name,
    into = c("first_name", "last_name"),
    sep = "\\s+",
    extra = "merge",
    fill = "right")

employees_clean_NW

## # A tibble: 5 × 6
##   id    first_name   last_name birthdate  gender salary
##   <chr> <chr>        <chr>     <date>     <fct>   <dbl>
## 1 001   John         Smith     1985-04-23 M       45000
## 2 002   Alice        Brown     1991-12-05 F       52000
## 3 003   Mark         Davis     1979-06-12 M          NA
## 4 004   SophieMartin <NA>      1988-02-28 F       61000
## 5 005   Paul         Durand    1993-05-17 M       48000

# Because:
# There is no separator
# separate() cannot infer a split point
# The entire string is assigned to first_name
# last_name becomes NA
# So again, the failure is not technical, but assumption-based.

# Go ahead for a pattern:
# First name starts with one uppercase letter, followed by lowercase letters
# Last name starts with one uppercase letter, followed by lowercase letters
# This pattern holds even when there is no space
# This allows us to detect the boundary between two CamelCase words.

# An elegant solution
employees_clean_elegant <- employees_clean %>%
  mutate(
    matches = str_match(name, "^([A-Z][a-z]+)\\s*([A-Z][a-z]+)$"),
    first_name = matches[, 2],
    last_name  = matches[, 3]
  ) %>% select(-matches, -name)

employees_clean_elegant

## # A tibble: 5 × 6
##   id    birthdate  gender salary first_name last_name
##   <chr> <date>     <fct>   <dbl> <chr>      <chr>
## 1 001   1985-04-23 M       45000 John       Smith
## 2 002   1991-12-05 F       52000 Alice      Brown
## 3 003   1979-06-12 M          NA Mark       Davis
## 4 004   1988-02-28 F       61000 Sophie     Martin
## 5 005   1993-05-17 M       48000 Paul       Durand
```

```r
# A simple one
employees_clean_simple <- employees_clean %>%
  mutate(
    first_name = str_extract(name, "^[A-Z][a-z]+"),
    last_name  = str_extract(name, "[A-Z][a-z]+$")
  ) %>%
  select(-name)

employees_clean_simple

## # A tibble: 5 × 6
##   id    birthdate  gender salary first_name last_name
##   <chr> <date>     <fct>   <dbl> <chr>      <chr>
## 1 001   1985-04-23 M       45000 John       Smith
## 2 002   1991-12-05 F       52000 Alice      Brown
## 3 003   1979-06-12 M          NA Mark       Davis
## 4 004   1988-02-28 F       61000 Sophie     Martin
## 5 005   1993-05-17 M       48000 Paul       Durand



#Exercise 9.4
FE_1307 = read_tsv("ForeignAid.tsv")

FE_1307$Country[FE_1307$Country %in% c("Korea–North", "Korea,N", "Korea-
North", "Korea North", "N Korea", "Korea, N", "Korea, No", "Korea, North",
"Korea::North", "Korea ;North", "Korea. No.", "Korea, Nor", "Korea north",
"Korea xNorth")] = "North Korea"
FE_1307$Country[FE_1307$Country %in% c("Korea South", "SouthKorea", "south
Korea", "So Korea", "Korea, South", "Korea: South", "Korea `South", "south
korea", "Korea, So;", "Korea-South", "Korea::South")] = "South Korea"
unique(FE_1307$Country)

## [1] "North Korea" "South Korea"

FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Dept of Agriculture",
"Department of Argic", "Department of Agriculture", "Department -
Agriculture", "Department of Agricult", "Department of agriculture")] =
"Agriculture"
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Department – State",
"Department of State", "department of State", "department of State", "State
Department", "Department of State", "Department Of State", "Department -
State")] = "State"
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("U.S. Agency for
International Development", "US Agency for International Development", "U.S.
Agency f/ International Development", "U.S. Agency: International
Development", "U.S. Agency for INt'l Development", "USA Agency for
International Development", "International Development")] = "International
Development"
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Trade and Development
Agency", "Department of Commerce", "Federal Trade Commission")] = "Commerce"
```

```r
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Department of Energy")] =
"Energy"
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Department- Health and
Human Services", "Department of Health and Human Services", "Department of
Health and Human Svc")] = "Health&Human"
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Department of Justice")]
= "Justice"
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Environmental Protection
Agency")] = "Environment"
FE_1307$FundingAgency[FE_1307$FundingAgency %in% c("Department of Defense",
"Department of Homeland Sec")] = "Defense"
unique(FE_1307$FundingAgency)
```

```
## [1] "Agriculture"                 "State"
## [3] "International Development"  "Commerce"
## [5] "Energy"                      "Health&Human"
## [7] "Justice"                     "Environment"
## [9] "Defense"
```

```r
agency = length(unique(FE_1307$FundingAgency))
agency
```

```
## [1] 9
```

```r
FE_1307$FundingAmount <-
lapply(FE_1307$FundingAmount,gsub,pattern="$",fixed=TRUE,replacement="")
FE_1307$FundingAmount <-
lapply(FE_1307$FundingAmount,gsub,pattern=",",fixed=TRUE,replacement="")
FE_1307$FundingAmount <-
lapply(FE_1307$FundingAmount,gsub,pattern="a",fixed=TRUE,replacement="")
FE_1307$FundingAmount <-lapply(FE_1307$FundingAmount,gsub,pattern="
",fixed=TRUE,replacement="")
FE_1307$FundingAmount <-lapply(FE_1307$FundingAmount,gsub,pattern="-
",fixed=TRUE,replacement="")

FE_1307 <- FE_1307 %>%
  mutate(FundingAmount=as.numeric(FundingAmount))

#summary(FE_1307)
#boxplot(FE_1307$FundingAmount)
#hist(FE_1307$FundingAmount)

table(FE_1307$FundingAgency)
```

```
##
##              Agriculture                  Commerce                   Defense
##                       21                         9                         4
##                   Energy               Environment              Health&Human
##                       13                         1                         4
## International Development                   Justice                     State
##                       29                         5                        41
```